Weekly ID + General Medicine Literature Digest (30-05-2026)

- Window: last 7 days
- Core digest items: 15
- Extended digest items: 25
- Total papers scored: 254
- LLM summary success: 100.0% (40/40)

Core Digest (10-15 mins)

1. **Fast Antimicrobial Susceptibility Testing for Gram-Negative Bacteremia: The FAST Randomized Clinical Trial.**

   Journal: JAMA | RCT | Date: 26-05-2026 | Score: 23 (rule 20, llm +3) | Translation horizon: 0-12 months

   PubMed: https://pubmed.ncbi.nlm.nih.gov/41999287/

   DOI: https://doi.org/10.1001/jama.2026.5487

   Trial n: n=850

   Why it matters:

   - Gram-negative bloodstream infections are associated with high morbidity and mortality, particularly in settings with high antimicrobial resistance.
   - Timely and appropriate antibiotic therapy is critical for improving outcomes in patients with gram-negative bacteremia.
   - Rapid antimicrobial susceptibility testing (AST) methods offer the potential to accelerate treatment decisions and optimize antibiotic stewardship.

   Headline result: Rapid phenotypic antimicrobial susceptibility testing was not superior to standard testing for the primary outcome of desirability of outcome ranking (DOOR) at day 30 (probability 48.8%, 95% CI 45.3%-52.4%) in patients with gram-negative bacilli bloodstream infections.

   Major limitation: The open-label design of the trial could introduce bias, particularly for a composite primary outcome that involves subjective assessment of "deleterious events."

   Clinical takeaway:

   - Do not routinely adopt rapid phenotypic AST for all gram-negative bacteremia cases based on these overall negative findings for the primary clinical outcome.
   - Consider the potential benefit of rapid AST in specific high-risk populations, such as those with suspected carbapenem-resistant infections, where it showed a trend towards faster effective therapy.
   - Continue to rely on robust antimicrobial stewardship programs to guide appropriate antibiotic selection and de-escalation.

   Read priority: Read if time

---

2. **Efficacy and safety of the CD40 ligand inhibitor dapirolizumab pegol in systemic lupus erythematosus (PHOENYCS GO): a randomised, double-blind, placebo-controlled, phase 3 trial.**

   Journal: Lancet | RCT | Date: 29-05-2026 | Score: 20 (rule 17, llm +3) | Translation horizon: >12 months

   PubMed: https://pubmed.ncbi.nlm.nih.gov/42214397/

   DOI: https://doi.org/10.1016/S0140-6736(26)00691-4

   Trial n: n=315 (full-analysis set)

   Why it matters:

   - Systemic lupus erythematosus (SLE) is a chronic autoimmune disease causing significant morbidity and reduced quality of life.
   - Many patients with moderate-to-severe active SLE do not achieve adequate disease control with existing standard-of-care therapies.
   - New therapeutic options are needed to improve clinical outcomes and address unmet needs in this patient population.

Headline result: A significantly greater proportion of patients receiving dapirolizumab pegol (50%) versus placebo (35%) had BICLA response at week 48 (difference 14.6%; 95% CI 3.3-25.8; p=0.011).

Major limitation: The trial duration of 48 weeks limits assessment of long-term safety and durability of response.

Clinical takeaway:

- Dapirolizumab pegol demonstrated significant improvement in disease activity for moderate-to-severe SLE patients.
- Clinicians should be aware of potential hypersensitivity reactions and monitor for thromboembolic events if this agent becomes available.
- Further studies are needed to confirm long-term safety and efficacy and to understand its role in the treatment algorithm.

Read priority: Read now

---

3. **Efficacy and safety of a 4-month quabodepistat, delamanid, and bedaquiline regimen for drug-susceptible pulmonary tuberculosis: a multicentre, open-label, randomised, proof-of-concept, non-inferiority, phase 2b/c trial.**

   Journal: Lancet Infect Dis | RCT | Date: 29-05-2026 | Score: 19 (rule 16, llm +3) | Translation horizon: >12 months

   PubMed: https://pubmed.ncbi.nlm.nih.gov/42214407/

   DOI: https://doi.org/10.1016/S1473-3099(26)00143-X

   Trial n: n=122 enrolled, 121 in mITT analysis set

   Why it matters:

   - Tuberculosis (TB) remains a leading cause of morbidity and mortality globally, requiring effective and accessible treatments.
   - The current standard 6-month regimen for drug-susceptible TB is lengthy, contributing to poor adherence and treatment failures.
   - Shorter, highly effective treatment regimens are critically needed to improve patient outcomes and enhance public health efforts against TB.

   Headline result: A 4-month regimen of quabodepistat, delamanid, and bedaquiline (DBQ) met non-inferiority to the 6-month standard of care (RHEZ) for sputum culture conversion (96.0% vs 100.0%, difference -4.0% [80% CI -7.4 to 3.4]) in drug-susceptible pulmonary tuberculosis.

   Major limitation: This was an open-label, phase 2b/c proof-of-concept trial with a relatively small sample size and a wide non-inferiority margin, limiting definitive conclusions.

   Clinical takeaway:

   - Monitor for results from larger, phase 3 trials to confirm the efficacy and safety of this promising 4-month regimen.
   - If confirmed, this regimen could significantly shorten treatment duration for drug-susceptible pulmonary tuberculosis, potentially improving adherence and outcomes.

   Read priority: Read if time

---

4. **Time to HIV rebound after infusion of long-acting broadly neutralising antibodies 3BNC117-LS and 10-1074-LS and analytical treatment interruption (the RIO trial): a double-blind, randomised, placebo-controlled trial.**

   Journal: Lancet HIV | RCT | Date: 27-05-2026 | Score: 17 (rule 14, llm +3) | Translation horizon: >12 months

   PubMed: https://pubmed.ncbi.nlm.nih.gov/42202839/

   DOI: https://doi.org/10.1016/S2352-3018(26)00059-7

   Trial n: n=68

   Why it matters:

   - Lifelong antiretroviral therapy (ART) for HIV presents challenges including adherence, side effects, and stigma, driving the search for ART-free remission strategies.
   - Broadly neutralising antibodies (bNAbs) represent a promising investigational approach to achieve sustained viral control without daily ART.

- Demonstrating the efficacy of long-acting bNAbs in maintaining ART-free viral suppression could significantly impact the future management and quality of life for people with HIV.

Headline result: Long-acting bNAbs (3BNC117-LS and 10-1074-LS) significantly reduced viral rebound, with 75% of participants in the bNAb arm remaining virally suppressed at 20 weeks post-ART interruption compared to 11% in the placebo arm (HR 0.09; 95% CI 0.04-0.21, p<0.0001).

Major limitation: The study population was highly selected (early-stage HIV, virally suppressed, no viral insensitivity to 10-1074), limiting the generalizability of these findings to broader HIV populations.

Clinical takeaway:

- These long-acting bNAbs show significant promise for achieving ART-free viral control in carefully selected individuals with early-stage HIV infection.
- Clinicians should watch for further research on bNAb combinations, optimal dosing, and patient selection criteria as this strategy moves towards broader clinical applicability.
- This study represents an important step towards HIV remission, but bNAb therapy is not yet ready for routine clinical use outside of research settings.

Read priority: Read now

---

5. **Switching to Daily Doravirine/Islatravir (100/0.25 mg) Maintains Viral Suppression Through Week 48 Despite Baseline Non-Nucleoside Reverse Transcriptase Inhibitor Resistance-Associated Mutations or M184I/V in Proviral DNA.**

Journal: J Infect Dis | Date: 29-05-2026 | Score: 16 (rule 13, llm +3) | Translation horizon: 0-12 months

PubMed: https://pubmed.ncbi.nlm.nih.gov/42213454/

DOI: https://doi.org/10.1093/infdis/jiag257

Trial n: n=1227

Why it matters:

- Doravirine/Islatravir (DOR/ISL) is a novel 2-drug regimen in development for HIV-1 treatment.
- Pre-existing resistance-associated mutations (RAMs) can complicate treatment selection and lead to virologic failure in people living with HIV.
- Understanding the efficacy of new regimens in the presence of common baseline RAMs is crucial for optimizing treatment strategies for virologically suppressed patients.

Headline result: At week 48, switching to daily DOR/ISL (100/0.25 mg) maintained HIV-1 RNA <50 copies/mL in ">=88%" of participants, with baseline NNRTI RAMs and/or M184I/V in proviral DNA not impacting virologic outcomes.

Major limitation: The study population consisted of virologically suppressed participants, limiting generalizability to treatment-naive or treatment-experienced patients with active viremia.

Clinical takeaway:

- Consider DOR/ISL as a potential switch option for virologically suppressed patients, even if baseline proviral DNA testing reveals NNRTI RAMs or M184I/V.
- The presence of NNRTI RAMs or M184I/V in proviral DNA at baseline may not preclude successful virologic suppression with DOR/ISL in switch settings.

Read priority: Read if time

---

6. **Characterization of the VHH-Fc construct rimteravimab in healthy adults and patients hospitalized for mild-to-moderate COVID-19: Two Phase 1 randomized clinical trials.**

Journal: PLoS Med | RCT | Date: 30-05-2026 | Score: 15 (rule 12, llm +3) | Translation horizon: >12 months

PubMed: https://pubmed.ncbi.nlm.nih.gov/42213671/

DOI: https://doi.org/10.1371/journal.pmed.1004609

Trial n: n=57 (30 healthy volunteers, 27 hospitalized patients)

Why it matters:

- VHH-Fc constructs represent an innovative class of antibodies with potential for targeted therapies against various pathogens.

- Rimteravimab (XVR011) was developed as a humanized antibody to target SARS-CoV-2, addressing the ongoing need for effective COVID-19 treatments.
- Understanding the safety and pharmacokinetic profile of novel antibody formats like VHH-Fc is crucial for their future clinical development and application in infectious diseases.

Headline result: In two Phase 1 trials, rimteravimab demonstrated a favorable safety, tolerability, pharmacokinetic, and immunogenicity profile in 30 healthy volunteers and 27 hospitalized COVID-19 patients, with no serious treatment-emergent adverse events in healthy volunteers and mostly mild-to-moderate events in patients.

Major limitation: The study was limited by its small sample size, the absence of a placebo arm in the patient study, and the loss of rimteravimab potency against emerging SARS-CoV-2 Omicron BA.2 variants.

Clinical takeaway:

- This Phase 1 study provides initial safety and pharmacokinetic data for a novel VHH-Fc construct, rimteravimab, but does not assess its efficacy against current SARS-CoV-2 variants.
- Further research is needed to develop VHH-Fc constructs with broader variant coverage and to evaluate their clinical utility in larger, placebo-controlled efficacy trials.

Read priority: Read if time

---

7. **Trends in AIDS-defining illnesses among people living with HIV in clinical care in Europe: results from the EuroSIDA multicentre cohort study (2003-24).**

Journal: Lancet HIV | Date: 26-05-2026 | Score: 14 (rule 11, llm +3) | Translation horizon: 0-12 months

PubMed: https://pubmed.ncbi.nlm.nih.gov/42190668/

DOI: https://doi.org/10.1016/S2352-3018(26)00052-4

Trial n: n=22,119

Why it matters:

- AIDS-defining illnesses (ADIs) remain a substantial cause of morbidity and mortality among people with HIV in Europe, despite advancements in antiretroviral therapy (ART).
- Understanding the temporal trends and associated risk factors for ADIs is crucial for informing public health strategies and clinical care.
- Significant inequalities in HIV care persist across European regions, leading to a disproportionate burden of AIDS in certain areas, particularly Eastern Europe.

Headline result: The overall incidence rate of initial AIDS-defining illnesses was 7.7 per 1000 person-years of follow-up, with the largest absolute decline observed in Eastern Europe, where the incidence rate fell from 115.1 to 15.7 per 1000 person-years between 2003 and 2013.

Major limitation: As an observational cohort study, it cannot establish causality and may be subject to unmeasured confounding.

Clinical takeaway:

- Maintain vigilance for AIDS-defining illnesses, especially opportunistic infections, in people living with HIV, even those on ART.
- Identify and closely monitor patients with risk factors such as injecting drug use, lower CD4 counts, or interrupted ART for increased risk of ADIs.
- Support public health initiatives focused on improving access to and adherence to ART, particularly in regions with a higher burden of ADIs like Eastern Europe.

Read priority: Read if time

---

8. **Early Discontinuation of Empiric Antibiotics in Pediatric Haploidentical Hematopoietic Cell Transplant Recipients with Febrile Neutropenia.**

Journal: Clin Infect Dis | Date: 26-05-2026 | Score: 14 (rule 11, llm +3) | Translation horizon: 0-12 months

PubMed: https://pubmed.ncbi.nlm.nih.gov/42189672/

DOI: https://doi.org/10.1093/cid/ciag327

Trial n: n=172 (82 Post-I, 90 Pre-I)

Why it matters:

- Febrile neutropenia is a common and serious complication in children, adolescents, and young adults (CAYA) undergoing haploidentical hematopoietic cell transplant (haplo-HCT).
- The optimal duration of empiric antibiotics for febrile neutropenia in this vulnerable population is not well established, leading to potential overuse.
- Reducing unnecessary antibiotic exposure in immunocompromised patients is crucial to mitigate antimicrobial resistance and drug-related adverse events.

Headline result: Early antibiotic discontinuation in CAYA haplo-HCT recipients with febrile neutropenia significantly reduced days of therapy by a median of 8 days ($p<0.01$) and was associated with a 70% (95% CI = 61, 78%) probability of a more favorable clinical outcome without increasing adverse events.

Major limitation: This was a retrospective, quasi-experimental, single-center study, which limits the generalizability and strength of evidence compared to a prospective randomized trial.

Clinical takeaway:

- Consider implementing a guideline for early empiric antibiotic discontinuation (within 72 hours) in CAYA haplo-HCT recipients with febrile neutropenia if no infection is identified, even with persistent fever.
- Monitor for results from larger, prospective multicenter trials to further validate these findings and support broader guideline changes.

Read priority: Read if time

---

9. **Pneumococcal membrane particles promote serotype-independent cellular and humoral immunity and protect against pneumococcal colonization.**

Journal: Proc Natl Acad Sci U S A | Date: 26-05-2026 | Score: 13 (rule 10, llm +3) | Translation horizon: >12 months

PubMed: https://pubmed.ncbi.nlm.nih.gov/42154558/

DOI: https://doi.org/10.1073/pnas.2537226123

Trial n: not reported

Why it matters:

- Pneumococcal infections remain a significant global health burden despite existing conjugated vaccines (PCVs).
- Current PCVs have serotype-specific limitations, necessitating novel strategies for broader protection.
- A vaccine capable of inducing serotype-independent immunity and preventing colonization could significantly enhance public health efforts against pneumococcal disease.

Headline result: Pneumococcal membrane particles (MP) promote serotype-independent cellular and humoral immunity in mice and humans, and immunization with MP alone protects against pneumococcal colonization in mice; a specific numeric effect size for protection was not reported.

Major limitation: This is a pre-clinical study conducted in mice and human cell lines, and its findings require validation in human clinical trials.

Clinical takeaway:

- Monitor ongoing research into pneumococcal membrane particles as a promising, serotype-independent vaccine candidate.
- Consider the potential for such a vaccine to not only prevent invasive disease but also reduce colonization, thereby limiting transmission.

Read priority: Awareness only

---

10. **Efficacy and safety of intravenous prasinezumab in individuals with early-stage Parkinson's disease on stable symptomatic monotherapy (PADOVA): a phase 2b, multicentre, randomised, double-blind, placebo-controlled study.**

Journal: Lancet | RCT | Date: 30-05-2026 | Score: 13 (rule 10, llm +3) | Translation horizon: >12 months

PubMed: https://pubmed.ncbi.nlm.nih.gov/42208564/

DOI: https://doi.org/10.1016/S0140-6736(26)00865-2

Trial n: n=586 (293 per group)

Why it matters:

- Parkinson's disease is a progressive neurodegenerative disorder with a significant unmet need for disease-modifying treatments.
- Targeting alpha-synuclein aggregation is a promising strategy to slow disease progression, addressing a core pathological mechanism.
- Effective therapies that delay motor progression could substantially improve long-term outcomes and quality of life for patients with early-stage Parkinson's disease.

Headline result: The primary endpoint was not met, with prasinezumab showing a non-significant delay in motor progression compared to placebo (hazard ratio 0.84 [95% CI 0.69-1.01]; p=0.066).

Major limitation: The trial did not meet its primary endpoint, indicating a lack of statistically significant efficacy for the chosen primary outcome.

Clinical takeaway:

- Prasinezumab did not demonstrate a statistically significant benefit in delaying motor progression in early-stage Parkinson's disease in this phase 2b trial.
- Clinicians should await results from the ongoing phase 3 PARAISO trial before considering prasinezumab for patient management.

Read priority: Awareness only

---

11. **Efficacy and safety of ocrelizumab in primary progressive multiple sclerosis, including older patients and those with more advanced disease (ORATORIO-HAND): a multicentre, double-blind, randomised, placebo-controlled, phase 3b study.**

Journal: Lancet | RCT | Date: 30-05-2026 | Score: 13 (rule 10, llm +3) | Translation horizon: 0-12 months

PubMed: https://pubmed.ncbi.nlm.nih.gov/42208561/

DOI: https://doi.org/10.1016/S0140-6736(26)00617-3

Trial n: n=1013 (ocrelizumab n=505; placebo n=508)

Why it matters:

- Primary progressive multiple sclerosis (PPMS) is a debilitating neurodegenerative disease with limited treatment options, making effective therapies crucial.
- Ocrelizumab is an approved treatment for PPMS, but its efficacy and safety in a broader population, including older patients and those with more advanced disease, needed further investigation.
- Clarifying the role of ocrelizumab in preserving hand function and overall disability progression in a wider PPMS patient group can significantly impact treatment decisions.

Headline result: Ocrelizumab reduced the risk of 12-week composite confirmed disability progression (12W-cCDP) by 30% (HR 0.70, 95% CI 0.57-0.86; p=0.0007) compared to placebo in a broad PPMS population.

Major limitation: The study is still ongoing, meaning long-term safety and efficacy data beyond the reported 144 weeks are not yet fully available.

Clinical takeaway:

- Consider ocrelizumab for a broader range of PPMS patients, including those who are older or have more advanced disease, given its demonstrated efficacy in this population.
- Ocrelizumab showed a stronger effect on hand function, which may be a key consideration for patients with specific functional deficits.
- Continue to monitor for infections, although the overall safety profile remained manageable and similar to placebo when excluding COVID-19 cases.

Read priority: Read now

---

12. **Efficacy and safety of oral semaglutide 14 mg (flexible dose) in early-stage symptomatic Alzheimer's disease (evoke and evoke+): two phase 3, randomised, placebo-controlled trials.**

Journal: Lancet | RCT | Date: 30-05-2026 | Score: 13 (rule 10, llm +3) | Translation horizon: 0-12 months

PubMed: https://pubmed.ncbi.nlm.nih.gov/41865758/

DOI: https://doi.org/10.1016/S0140-6736(26)00459-9

Trial n: n=3808

Why it matters:

- Alzheimer's disease remains a devastating condition with a critical need for effective disease-modifying therapies.
- Preclinical and observational data suggested GLP-1 receptor agonists might reduce dementia risk, leading to high anticipation for this class of drugs in AD.
- A safe and effective oral treatment for early Alzheimer's disease would represent a major therapeutic breakthrough, impacting millions globally.

Headline result: In two phase 3 trials (evoke and evoke+), oral semaglutide 14 mg did not significantly slow clinical progression in early Alzheimer's disease, with estimated differences in CDR-SB score of -0.08 (95% CI -0.35 to 0.20, p=0.57) in evoke and 0.10 (95% CI -0.17 to 0.38, p=0.46) in evoke+ compared to placebo.

Major limitation: The early discontinuation of the trials due to futility, while appropriate, means that any extremely subtle or very long-term benefits of semaglutide could not be fully assessed.

Clinical takeaway:

- Do not consider oral semaglutide for the treatment or prevention of cognitive decline in patients with early Alzheimer's disease.
- These results provide a definitive answer regarding semaglutide's efficacy for this indication, closing a highly anticipated line of inquiry for this specific drug.
- Further research into other GLP-1 receptor agonists or different mechanisms for Alzheimer's disease should continue, as this class has shown promise in other neurological contexts.

Read priority: Read now

---

13. **Tirzepatide Versus Intensified Conventional Care After 2 Years of Treatment in Early Type 2 Diabetes : A Randomized Clinical Trial.**

Journal: Ann Intern Med | RCT | Date: 26-05-2026 | Score: 13 (rule 10, llm +3) | Translation horizon: 0-12 months

PubMed: https://pubmed.ncbi.nlm.nih.gov/42184419/

DOI: https://doi.org/10.7326/ANNALS-25-05602

Trial n: n=794

Why it matters:

- Early and sustained glycemic control in type 2 diabetes (T2D) is critical for preventing long-term complications and improving patient outcomes.
- Tirzepatide, a dual GIP/GLP-1 receptor agonist, offers a potent mechanism for glucose lowering and weight management, which may be particularly beneficial in early disease.
- Clinicians need robust comparative data on newer agents like tirzepatide against intensified conventional care to guide treatment decisions for patients with early T2D.

Headline result: Tirzepatide was superior to intensified conventional care (ICC) for mean HbA1c reduction at 2 years (-1.99 percentage points vs -1.32 percentage points; estimated treatment difference -0.68 percentage points; $P < 0.001$).

Major limitation: The open-label design of the study could introduce bias.

Clinical takeaway:

- Consider tirzepatide for patients with early type 2 diabetes uncontrolled on metformin, especially those who may benefit from significant HbA1c, weight, and waist circumference reductions.
- Monitor for gastrointestinal adverse events, which were common in both treatment groups.
- Discuss the potential for achieving normoglycemia (HbA1c <5.7%) with tirzepatide, as 60.2% of participants achieved this compared to 24.0% with ICC.

Read priority: Read now

---

14. **Safety and efficacy of astegolimab for COPD with frequent exacerbations regardless of baseline blood eosinophil counts (ALIENTO and ARNASA): randomised, double-blind, placebo-controlled, phase 2b and 3 trials.**

Journal: Lancet | RCT | Date: 23-05-2026 | Score: 13 (rule 10, llm +3) | Translation horizon: 0-12 months

PubMed: https://pubmed.ncbi.nlm.nih.gov/42150581/

DOI: https://doi.org/10.1016/S0140-6736(26)00637-9

Trial n: n=2676 across two trials (ALIENTO n=1301, ARNASA n=1375)

Why it matters:

- COPD exacerbations significantly contribute to morbidity, mortality, and healthcare burden for patients.
- Current therapeutic options for preventing frequent COPD exacerbations, particularly in patients irrespective of baseline blood eosinophil counts, remain limited.
- Targeting the IL-33/ST2 pathway represents a novel mechanistic approach to reduce exacerbation frequency in a broad population of COPD patients.

Headline result: In the ALIENTO trial, astegolimab every 2 weeks reduced the annual rate of moderate or severe COPD exacerbations by 15% (rate ratio 0.85, 95% CI 0.72-1.00; p=0.049) compared to placebo, while the ARNASA trial did not meet statistical significance for its primary endpoint.

Major limitation: The primary endpoint was only statistically significant in one of the two pivotal trials (ALIENTO) for the every 2-week dosing, and not in the other (ARNASA), suggesting inconsistent efficacy.

Clinical takeaway:

- Watch for further data on astegolimab, particularly regarding its efficacy across different COPD phenotypes and the consistency of its effect.
- Consider the potential role of IL-33/ST2 pathway inhibitors for patients with frequent COPD exacerbations who have limited treatment options, especially if not eosinophilic.

Read priority: Read if time

---

15. **Growth in confinement promotes Pseudomonas aeruginosa tolerance to antibiotics.**

    Journal: Proc Natl Acad Sci U S A | Date: 02-06-2026 | Score: 12 (rule 9, llm +3) | Translation horizon: >12 months

    PubMed: https://pubmed.ncbi.nlm.nih.gov/42213750/

    DOI: https://doi.org/10.1073/pnas.2605176123

    Trial n: not reported

    Why it matters:

    - Pseudomonas aeruginosa is a significant pathogen often associated with chronic infections and biofilm formation in confined host environments.
    - Antibiotic tolerance, distinct from genetic resistance, can lead to treatment failures and persistent infections, posing a major clinical challenge.
    - Understanding novel mechanisms that contribute to antibiotic tolerance is crucial for developing more effective therapeutic strategies against P. aeruginosa.

    Headline result: In an in vitro study, Pseudomonas aeruginosa grown in confined elastic materials exhibited reduced sensitivity to antibiotics (colistin and tobramycin) in a stiffness-dependent manner, with active efflux and membrane remodeling identified as key regulators of this mechanically induced tolerance; no specific numeric outcome was reported.

    Major limitation: This was an in vitro study using synthetic hydrogels, which may not fully recapitulate the complex physiological conditions and host-pathogen interactions found in vivo.

    Clinical takeaway:

    - Consider that mechanical forces within host tissues could potentially contribute to antibiotic tolerance in P. aeruginosa infections, particularly in chronic or biofilm-associated settings.
    - Future research is needed to translate these in vitro findings into in vivo models and explore strategies to overcome mechanically induced tolerance to improve antibiotic efficacy.

    Read priority: Awareness only

Extended Digest (up to 60 minutes)

1. **Near-point-of-care diagnosis of tuberculosis with LyocartE MTB Assay: a multicentre diagnostic accuracy study.** Journal: Clin Infect Dis | Date: 29-05-2026 | Score: 11 (rule 9, llm +2) PubMed: https://pubmed.ncbi.nlm.nih.gov/42212698/ DOI: https://doi.org/10.1093/cid/ciag342 LLM note: The LyocartE assay demonstrated high diagnostic accuracy for tuberculosis from sputum and tongue swabs, comparable to Xpert Ultra, offering a rapid near point-of-care diagnostic tool. Read priority: Read now

---

2. **Performance of five mpox antigen-based rapid diagnostic tests tested on lesion swabs from patients with suspected mpox from the Kinshasa province of DR Congo: a diagnostic accuracy study.** Journal: Lancet Infect Dis | Date: 25-05-2026 | Score: 11 (rule 9, llm +2) PubMed: https://pubmed.ncbi.nlm.nih.gov/42184813/ DOI: https://doi.org/10.1016/S1473-3099(26)00141-6 LLM note: A diagnostic accuracy study of five mpox antigen-based rapid diagnostic tests on lesion swabs found the best performing assay had 77.3% sensitivity and 93.5% specificity, useful for screening in high-prevalence settings. Read priority: Read now

---

3. **Phase 3 Results of Bepirovirsen Treatment for Chronic Hepatitis B Virus Infection.** Journal: N Engl J Med | RCT | Date: 28-05-2026 | Score: 11 (rule 9, llm +2) PubMed: https://pubmed.ncbi.nlm.nih.gov/42206582/ DOI: https://doi.org/10.1056/NEJMoa2515131 LLM note: Two phase 3 randomized controlled trials showed that bepirovirsen significantly increased functional cure rates, defined by HBsAg loss and undetectable HBV DNA, in patients with chronic hepatitis B virus infection. Read priority: Read now

---

4. **Management of Chronic Subdural Hematoma With Adjunctive Embolization of Middle Meningeal Artery: The EMMA-Can Randomized Clinical Trial.** Journal: JAMA | RCT | Date: 26-05-2026 | Score: 11 (rule 9, llm +2) PubMed: https://pubmed.ncbi.nlm.nih.gov/42060283/ DOI: https://doi.org/10.1001/jama.2026.4910 LLM note: Adjunctive middle meningeal artery embolization after surgical drainage significantly reduced symptomatic recurrence of chronic subdural hematoma at 90 days compared to surgery alone in a randomized clinical trial. Read priority: Read now

---

5. **Burden and clinical impact of 'neglected' transfusion-transmitted infections in Cameroon: A systematic review and meta-analysis.** Journal: PLoS Negl Trop Dis | Date: 30-05-2026 | Score: 10 (rule 9, llm +1) PubMed: https://pubmed.ncbi.nlm.nih.gov/42189856/ DOI: https://doi.org/10.1371/journal.pntd.0014378 LLM note: A meta-analysis in Cameroon revealed a high prevalence of 'neglected' transfusion-transmitted infections, particularly Plasmodium spp. (16.6%), highlighting the urgent need for more epidemiological evidence and improved blood safety guidelines. Read priority: Read if time

---

6. **Clinical Impact and Cost-Effectiveness of Improving Access to Cryptococcal Meningitis Diagnostics and Treatment in Malawi.** Journal: Clin Infect Dis | Date: 28-05-2026 | Score: 10 (rule 8, llm +2) PubMed: https://pubmed.ncbi.nlm.nih.gov/42206850/ DOI: https://doi.org/10.1093/cid/ciag270 LLM note: Improving access to liposomal amphotericin B/flucytosine/fluconazole and expanding diagnostics for asymptomatic cryptococcal meningitis are cost-effective strategies to improve clinical outcomes in Malawi and similar settings. Read priority: Read now

---

7. **Exposure to antibiotics with anaerobe coverage in later life is associated with higher enteric pathobiont carriage.** Journal: J Infect | Date: 27-05-2026 | Score: 10 (rule 8, llm +2) PubMed: https://pubmed.ncbi.nlm.nih.gov/42208810/ DOI: https://doi.org/10.1016/j.jinf.2026.106774 LLM note: Exposure to antibiotics with anaerobe coverage, such as amoxicillin-clavulanate, is associated with increased enteric pathobiont carriage in long-term care residents, highlighting the importance of considering off-target commensal disruption. Read priority: Read now

---

8. **Upsurge of pneumococcal clade I-alpha/CC180 serotype 3 and its association with a LytA mutation linked to immune evasion and disease potential: an observational and experimental study.** Journal: Lancet Microbe | Date: 27-05-2026 | Score: 10 (rule 8, llm +2) PubMed: https://pubmed.ncbi.nlm.nih.gov/42202845/ DOI: https://doi.org/10.1016/j.lanmic.2026.101365 LLM note:

An emerging hypervirulent Streptococcus pneumoniae serotype 3 lineage (clade I-alpha/CC180) with a LytA mutation is associated with increased invasive pneumococcal disease and vaccine failures, highlighting a threat to vaccine effectiveness. Read priority: Read now

---

9. **Early adjunct anti-PD-L1 immunotherapy improves outcomes and restores infection-induced immune paralysis in mice with invasive pulmonary mucormycosis.** Journal: Proc Natl Acad Sci U S A | Date: 26-05-2026 | Score: 10 (rule 8, llm +2) PubMed: https://pubmed.ncbi.nlm.nih.gov/42160348/ DOI: https://doi.org/10.1073/pnas.2512042123 LLM note: Early adjunct anti-PD-L1 immunotherapy significantly improves outcomes and reverses immune paralysis in mice with invasive pulmonary mucormycosis, suggesting a promising host-directed strategy for severe mold infections. Read priority: Read now

---

10. **Anti-inflammatory effects of oral NLRP3 inhibition with ruvonoflast among individuals at elevated cardiovascular risk.** Journal: J Am Coll Cardiol | RCT | Date: 26-05-2026 | Score: 10 (rule 8, llm +2) PubMed: https://pubmed.ncbi.nlm.nih.gov/42187339/ DOI: https://doi.org/10.1016/j.jacc.2026.05.014 LLM note: Oral NLRP3 inhibitor ruvonoflast significantly reduced hsCRP and other inflammatory biomarkers in individuals with elevated cardiovascular risk, demonstrating anti-inflammatory efficacy in a Phase 1b trial. Read priority: Read now

---

11. **The Veterans Affairs' Whole Health Approach for Chronic Pain Management: The wHOPE Randomized Clinical Trial.** Journal: JAMA | Date: 26-05-2026 | Score: 10 (rule 9, llm +1) PubMed: https://pubmed.ncbi.nlm.nih.gov/42054020/ DOI: https://doi.org/10.1001/jama.2026.5006 LLM note: A whole health team approach for chronic pain in VA patients showed a statistically significant but small improvement in pain interference at 12 months compared to cognitive behavioral therapy or usual care. Read priority: Read if time

---

12. **Cross-kingdom metabolic interactions govern Candida albicans overgrowth and colitis progression.** Journal: Cell Host Microbe | Date: 28-05-2026 | Score: 9 (rule 8, llm +1) PubMed: https://pubmed.ncbi.nlm.nih.gov/42208527/ DOI: https://doi.org/10.1016/j.chom.2026.04.020 LLM note: Cross-kingdom metabolic interactions, involving Cladosporium tenuissimum limiting ornithine and Bacteroides fragilis providing threonine, govern Candida albicans overgrowth and colitis progression, with dietary threonine restriction attenuating colitis in mice. Read priority: Read if time

---

13. **Affinity-matured B cell responses neutralizing type-I interferons underlie severe viral infections.** Journal: Cell | Date: 28-05-2026 | Score: 9 (rule 8, llm +1) PubMed: https://pubmed.ncbi.nlm.nih.gov/42097138/ DOI: https://doi.org/10.1016/j.cell.2026.04.013 LLM note: Autoantibodies neutralizing type-I interferons, arising from affinity-matured memory B cell responses established before infection, are a core mechanism underlying severe viral diseases like COVID-19. Read priority: Read if time

---

14. **A deep mutational scanning-informed protein language model predicts SARS-CoV-2 evolution dynamics with spatiotemporal resolution.** Journal: Nat Microbiol | Date: 27-05-2026 | Score: 9 (rule 8, llm +1) PubMed: https://pubmed.ncbi.nlm.nih.gov/42204343/ DOI: https://doi.org/10.1038/s41564-026-02377-5 LLM note: DeepCoV, a deep-learning framework, accurately forecasts the dominance of emerging SARS-CoV-2 variants a month in advance, offering a scalable tool for real-time surveillance and public health responses. Read priority: Read if time

---

15. **Clinical and genomic characteristics of hypervirulent Klebsiella pneumoniae isolated from throat culture suggesting an emerging causative agent of pharyngitis.** Journal: BMC Infect Dis

| Date: 26-05-2026 | Score: 9 (rule 8, llm +1) PubMed: https://pubmed.ncbi.nlm.nih.gov/42192319/ DOI: https://doi.org/10.1186/s12879-026-13654-3 LLM note: Hypervirulent Klebsiella pneumoniae (hvKp) is identified as a potential causative agent of pharyngitis, with specific strains showing resistance genes despite phenotypic susceptibility to common antibiotics. Read priority: Read if time

---

16. **Amplification-free dual-blocking autocatalytic CRISPR-Cascade for attomolar DNA detection with low nonspecific signal.** Journal: Proc Natl Acad Sci U S A | Date: 26-05-2026 | Score: 9 (rule 8, llm +1) PubMed: https://pubmed.ncbi.nlm.nih.gov/42166245/ DOI: https://doi.org/10.1073/pnas.2537414123 LLM note: A novel dual-blocking CRISPR-Cascade system achieves amplification-free, attomolar DNA detection with high specificity and rapid results, demonstrating potential for decentralized diagnostics like MRSA detection. Read priority: Read if time

---

17. **Detecting changepoints in dynamical systems: Modeling time-varying transmission of seasonal influenza.** Journal: Proc Natl Acad Sci U S A | Date: 26-05-2026 | Score: 9 (rule 8, llm +1) PubMed: https://pubmed.ncbi.nlm.nih.gov/42154560/ DOI: https://doi.org/10.1073/pnas.2533861123 LLM note: A new framework accurately detects changepoints in seasonal influenza transmission rates, identifying consistent patterns during periods of increased social mixing and enabling improved epidemic forecasting. Read priority: Read if time

---

18. **Long-term antibody dynamics challenge the paradigm of lifelong homotypic immunity to dengue virus.** Journal: Proc Natl Acad Sci U S A | Date: 02-06-2026 | Score: 9 (rule 7, llm +2) PubMed: https://pubmed.ncbi.nlm.nih.gov/42201979/ DOI: https://doi.org/10.1073/pnas.2606206123 LLM note: Long-term follow-up data reveal that dengue virus antibody titers decay over years, leading to frequent homotypic reinfections in endemic settings and challenging the paradigm of lifelong immunity. Read priority: Read now

---

19. **Decompression with or without Duraplasty for Chiari I and Syringomyelia.** Journal: N Engl J Med | RCT | Date: 28-05-2026 | Score: 9 (rule 9) PubMed: https://pubmed.ncbi.nlm.nih.gov/42202320/ DOI: https://doi.org/10.1056/NEJMoa2402821 LLM note: A cluster-randomized trial in children with Chiari I malformation and syringomyelia found no significant difference in surgical complications between posterior fossa decompression with or without duraplasty, despite differences in syrinx reduction. Read priority: Awareness only

---

20. **Zanidatamab with and without Tislelizumab in HER2-Positive Gastroesophageal Cancer.** Journal: N Engl J Med | RCT | Date: 28-05-2026 | Score: 9 (rule 9) PubMed: https://pubmed.ncbi.nlm.nih.gov/42202319/ DOI: https://doi.org/10.1056/NEJMoa2517729 LLM note: A phase 3 randomized controlled trial demonstrated that zanidatamab plus chemotherapy, with or without tislelizumab, improved progression-free survival in HER2-positive advanced gastroesophageal adenocarcinoma compared to trastuzumab plus chemotherapy. Read priority: Awareness only

---

21. **Meiosis-specific genes play roles in ploidy reduction in Cryptococcus neoformans titan cells.** Journal: Proc Natl Acad Sci U S A | Date: 02-06-2026 | Score: 8 (rule 8) PubMed: https://pubmed.ncbi.nlm.nih.gov/42189998/ DOI: https://doi.org/10.1073/pnas.2522069123 LLM note: Meiosis-specific genes are critical for stable genome inheritance and ploidy reduction in Cryptococcus neoformans titan cells, impacting their ability to produce daughter cells and maintain genomic stability. Read priority: Awareness only

---

22. **Molecular basis of type VI secretion system effector loading.** Journal: Nat Microbiol | Date: 27-05-2026 | Score: 8 (rule 8) PubMed: https://pubmed.ncbi.nlm.nih.gov/42204345/ DOI:

https://doi.org/10.1038/s41564-026-02363-x LLM note: This study elucidates the molecular mechanism by which bacterial type VI secretion systems load effectors, revealing a stepwise process involving Hcp ring assemblies. Read priority: Awareness only

---

23. **A mechanism for substrate quality control in outer membrane protein assembly.** Journal: Proc Natl Acad Sci U S A | Date: 26-05-2026 | Score: 8 (rule 8) PubMed: https://pubmed.ncbi.nlm.nih.gov/42160346/ DOI: https://doi.org/10.1073/pnas.2536912123 LLM note: BamD-mediated regulation of BamA ensures substrate quality control during outer membrane protein assembly in Gram-negative bacteria, preventing accumulation of defective proteins and maintaining membrane integrity. Read priority: Awareness only

---

24. **Transcription start site heterogeneity controls MDA5 sensing of unspliced HIV-1 RNAs.** Journal: Proc Natl Acad Sci U S A | Date: 26-05-2026 | Score: 8 (rule 8) PubMed: https://pubmed.ncbi.nlm.nih.gov/42154552/ DOI: https://doi.org/10.1073/pnas.2522948123 LLM note: HIV-1 utilizes transcription start site heterogeneity to control MDA5 sensing of unspliced viral RNAs, allowing specific RNAs destined for packaging to be targeted while efficiently translated RNAs remain immunologically silent. Read priority: Awareness only

---

25. **A high-resolution, US-scale digital similar of interacting livestock, wild birds, and human ecosystems for multihost epidemic spread.** Journal: Proc Natl Acad Sci U S A | Date: 02-06-2026 | Score: 8 (rule 7, llm +1) PubMed: https://pubmed.ncbi.nlm.nih.gov/42207910/ DOI: https://doi.org/10.1073/pnas.2507074123 LLM note: The FIELD framework provides a high-resolution digital model of interacting livestock, wild birds, and human ecosystems to predict multihost epidemic spread, identifying H5N1 hotspots and informing surveillance. Read priority: Read if time

---